

Where the data comes from and how parity is computed

One page: sources, formulas, assumptions, cross-checks and limits. Press Ctrl/Cmd + P to print.

1 · SOURCES AND SERIES

Region	Numerator (meat)	Denominator (feed)	Source · Period
ABD	Slaughter cattle PPI WPU0131	Corn PPI WPU012202	FRED / BLS · 1971-01 → 2026-06 · 666 months
AB	Young bull carcass R3 EUR/100 kg	Feed grain (maize) EUR/tonne	European Commission Agri-food Data Portal · 2015-11 → 2026-07 · 129 months
Türkiye	Meat & meat products PPI TP.TUFE1YI.T17	Compound animal feed PPI TP.TUFE1YI.T25	CBRT EVDS / TurkStat · 2010-01 → 2026-06 · 198 months

The data is fetched automatically: a GitHub Actions job queries the sources monthly and writes the result to `/data/*.json`. The page, the film and the reports all read the same file — one single source of truth across three products. The EU series is a monthly mean of weekly quotations.

2 · PARITY AND THE INDEX

$$\text{parity}_{\text{month}} = \text{meat price}_{\text{month}} \div \text{feed price}_{\text{month}}$$

Parity is a unitless ratio: **how many units of feed one unit of meat buys**. Rising parity favours the feeder, falling parity hurts. Because the three regions use different measurement units, levels are not directly comparable; each series is therefore divided by its own 2016 mean and set to 100:

$$\text{index}_{\text{month}} = \text{parity}_{\text{month}} \div \text{parity}_{2016 \text{ mean}} \times 100$$

Why 2016? It is the first year with a full twelve months for all three series (EU data starts in November 2015). It is also a neutral year, close to each series' own long-run mean (EU 101, TR 104). **The base choice does not change the conclusion** — robustness check:

Base	ABD	AB	TR
2016 = 100 (used)	184	125	121
2015 = 100	162	123	120
Common-period mean (2016-2026)	187	126	127

3 · THE FEEDER P&L MODEL

The P&L scene in the film and on the page is a simple cost model. Per 100 units of sales revenue:

$$\begin{aligned} \text{feed cost} &= 35 \times (\text{long-run mean parity} \div \text{parity that month}) \\ \text{profit/loss} &= 100 - 45 \text{ (calf)} - 15 \text{ (other)} - \text{feed cost} \end{aligned}$$

What is data: the long-run means — US 1.08 · EU 2.24 · TR 1.04 — and each month's parity. **What is assumption:** the 45 / 35 / 15 shares, a calibration taken from typical ranges in sector feasibility studies. The logic: if parity doubles, the same animal's revenue buys twice the feed, so feed's share of revenue halves.

Not in the model: volatility of the replacement-calf price (which hit records in 2025-26), financing costs, currency risk, subsidy payments, taxes. The model is therefore a tool for **comparing periods**, not an absolute profit figure.

4 · STATISTICS

Lagged correlation: year-over-year changes are shifted ± 12 months; the lag with the highest correlation is read as the “time difference”. For US–EU, $r = 0.88$ at a lag of **0 months**. TR changes lead the US by 4 and the EU by 3 months.

Common pulse: principal component analysis (PC1) is applied to the standardised year-over-year changes of the three series; the variance share explained by the first component is reported as the “common wave”. In addition, a $TR \sim US + EU$ regression gives $R^2 = 0.50$.

Volatility: standard deviation over the mean for the last decade — US 34%, EU 21%, TR 19%. **Cycle alignment:** each country’s own trough month is set to 100 and subsequent months are expressed as ratios (trough = floor, ceiling ÷ floor = the coefficient).

5 · CROSS-CHECKS

Check	Our value	Independent source
US cattle PPI · Feb 2026	380,22	BLS: 380.217 — exact match
EU young bull R3 · May 2026	661,3 €	European Commission: €661.94 — within 0.1%
US cattle finishing · Dec 25–Mar 26	Model shows a profit	farmdoc: actual losses — the gap comes from holding the calf price fixed

6 · LIMITATIONS

- The ratios are not identical:** the US measures raw corn, the EU feed grain, TR compound feed; on the numerator side the US measures live animals, the EU carcasses, TR processed meat. Hence **changes are compared, not levels**.
- Series lengths differ:** US 55 years, TR 16, EU 11. “Long-run mean” therefore covers a different era in each country; the EU mean over-represents shock years.
- A PPI is not a farm-gate price:** the TR and US series are indices, not absolute lira or dollar prices.
- Quality and weight classes are held constant;** shifts in carcass weight and breed composition over time are not modelled.
- No causal claim is made.** Lagged correlation shows timing, not cause.

7 · SOURCE FILES

Raw data and code are public — every number can be verified:

- data/cattle-us.json · data/cattle-eu.json · data/cattle-tr.json
- Fetch code: `scripts/fetch_cattle_data.py` · Workflow: `.github/workflows/cattle-data.yml`
- Products: [interactive page](#) · [the film](#)